Distributed EV charging network

Hybrid MPI and OpenMP simulation

Three variants. 30 grid sizes. 30 line sizes. CAAS verification.

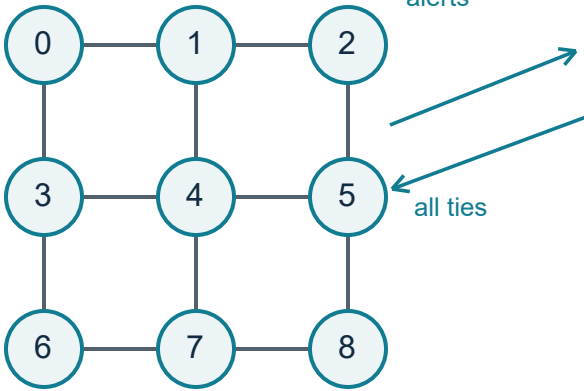
A measured study of neighbour communication and base-station congestion handling.

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Preparation used Codex. Full declaration and prompt records accompany this document.

One station process, one thread per port



alerts

all ties

Base station: world rank 0

Station i: world rank i+1

OpenMP team (k=4 shown)

T0 **T1** **T2** **T3**

Shared ports: [1, 0, 1, 0]

Only the initial thread calls MPI. Barriers publish each complete port snapshot.

MPI Cartesian stations 0..N-1

No wraparound. Missing edges: MPI_PROC_NULL.

Congestion triggers neighbour queries, then alerts

```
broadcast CONTINUE
threads update ports[i]
barrier; count free ports
exchange query intents with neighbours
reply only to requesting neighbours
if free < threshold:
    if neighbour has capacity: redirect
    else: send packed alert; receive advice
```

Query intent: 1 integer per real edge.

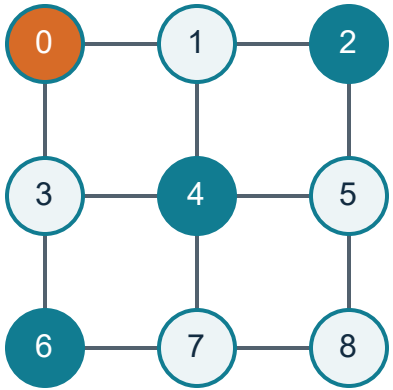
Report: ID, coordinates, UTC timestamp, capacities, counts and bytes.

Snapshot safety: one writer per port, atomic count, OpenMP barriers.

Tests: isolated node, full network, empty network and all ties.

Persistent threads avoid creating a new team for each simulation cycle.

The base returns every inferred nearest candidate



Orange: reporting station 0

Teal: equal-distance nodes 2, 4, 6

Receive the full batch, then update the cache.

Search the two-hop frontier in the Task 2 example. Manhattan distance = 2.

Exclude recent self-alerts and fresh low-capacity observations.

Return all remaining ties, or an empty list. Silence permits inferred availability.

Expiry is configurable. No reservation or guaranteed capacity is claimed.

Three versions preserve the same decisions

Version	Neighbour exchanges	Base communication	Candidate work
V0	Blocking Sendrecv	Rank-order Recv Blocking Send	Scan N nodes
V1	Irecv / Isend Waitall	Probe ANY_SOURCE Blocking Send	8 frontier offsets
V2	Irecv / Isend Waitall	Preposted Irecv Batched Isend	8 frontier offsets

Selection work per alert falls from O(N) to O(1). Nonblocking operations allow concurrency, while Waitall guarantees completion.

Timers measure local elapsed communication costs

Local study: two TCP-connected containers on one shared 16-CPU host.

30 grid sizes and 30 line sizes x 3 variants x 3 launches = 540 runs.

Two warmup cycles, six measured cycles, two port threads, mixed congestion.

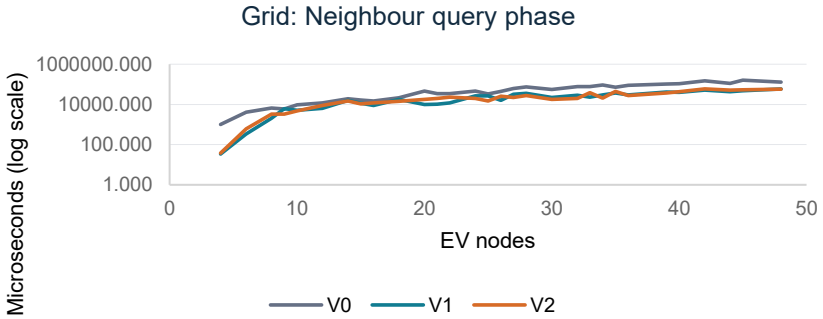
MPI_Wtime starts and stops on the same rank. No cross-host clock subtraction.

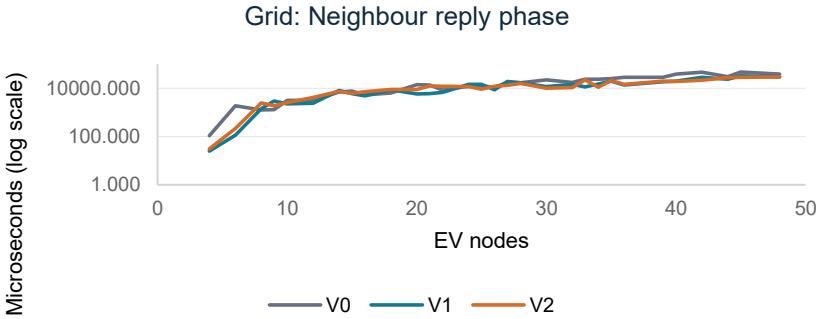
Query/reply: phase completion. Alert send: MPI completion. Redirect: receive wait.

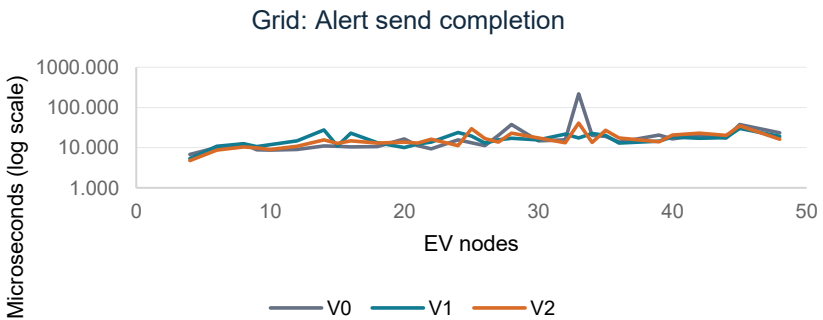
Round trip includes queueing and base computation. Raw rank-level records remain available.

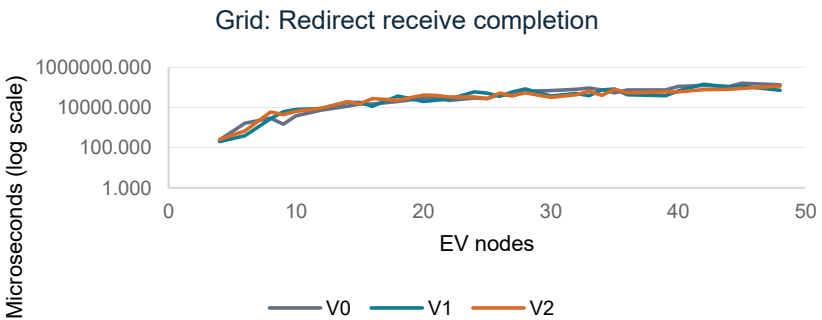
CAAS job 199603: 90 scaling runs, 5 grids + 5 lines, two allocated compute nodes.

Local grid scaling: bounded messages, increasing load



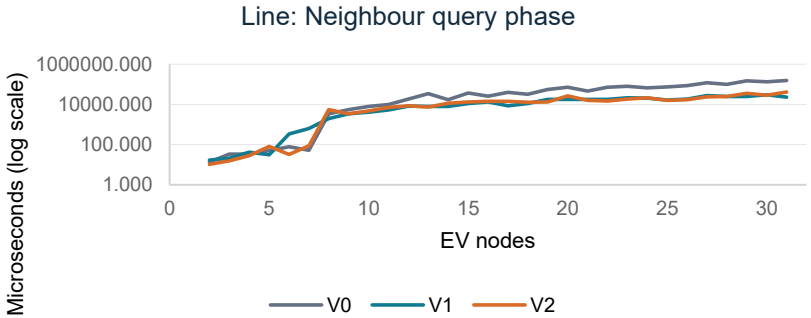


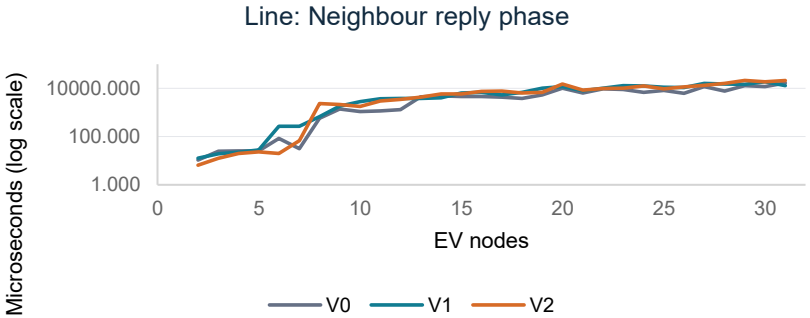


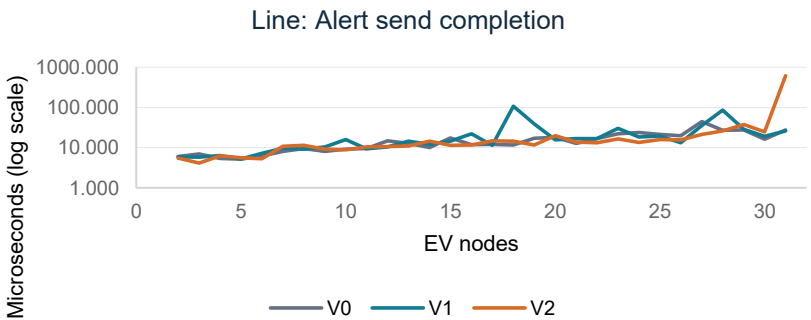


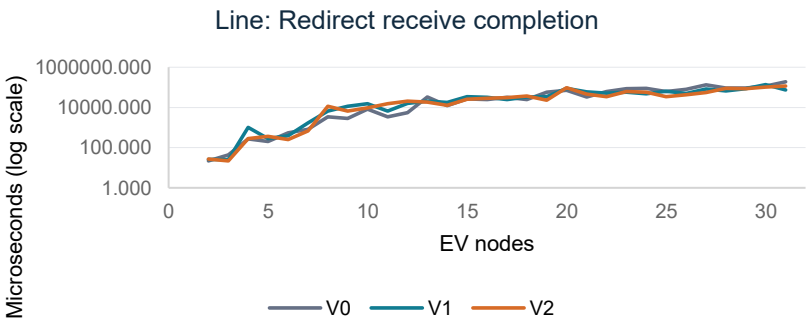
Selected V2, query: first five sizes 3312.2 us, last five 52526.6 us. Full ranges in appendix/CSV.

Local line scaling: bounded messages, increasing load





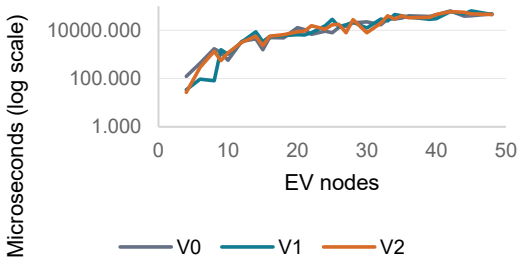




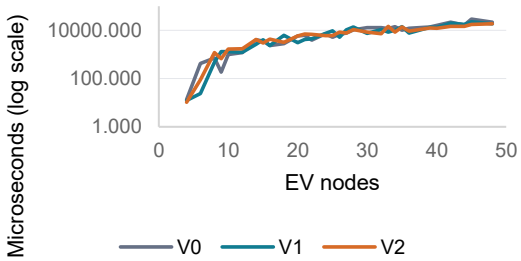
Selected V2, query: first five sizes 27.7 us, last five 28466.8 us. Full ranges in appendix/CSV.

Local control traffic and measured delay

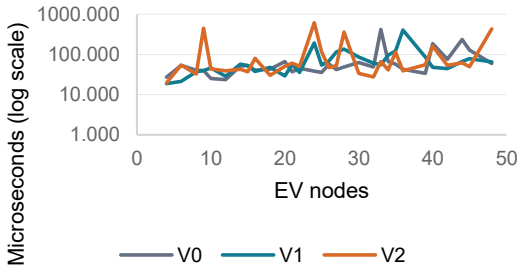
Grid: Control broadcast



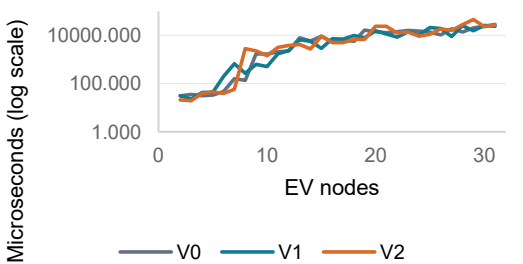
Grid: Alert-flag gather



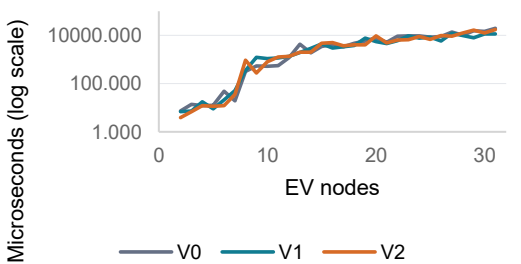
Grid: Shutdown ACK



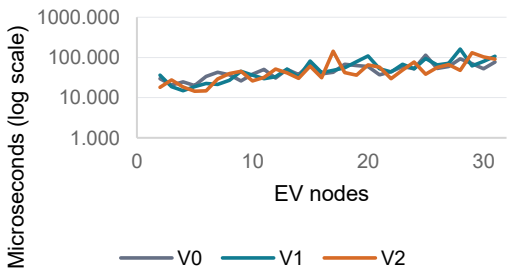
Line: Control broadcast



Line: Alert-flag gather

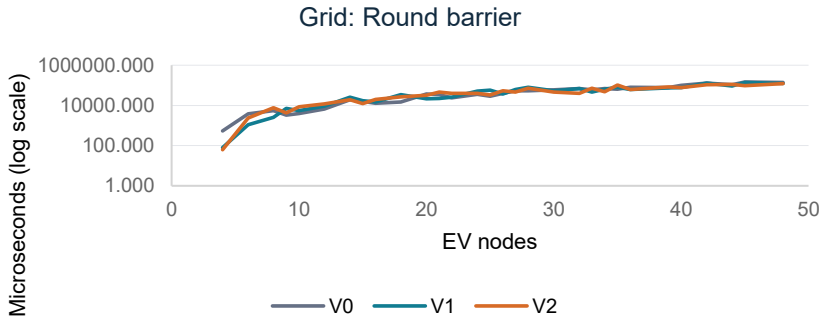


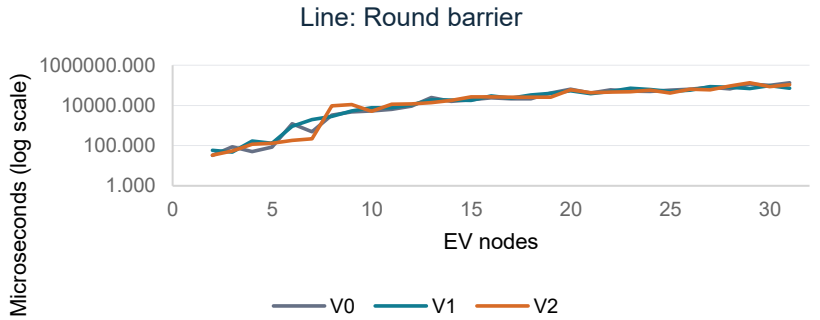
Line: Shutdown ACK

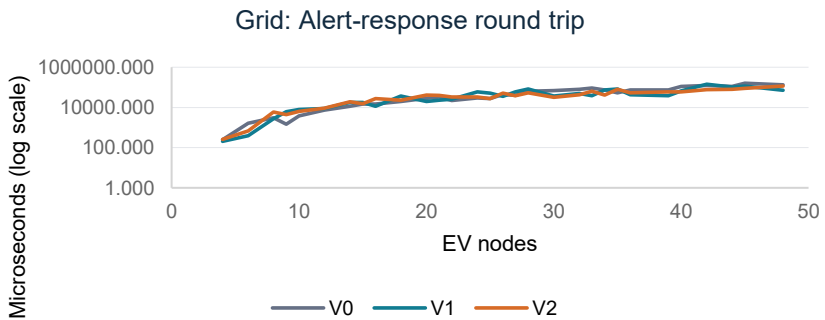


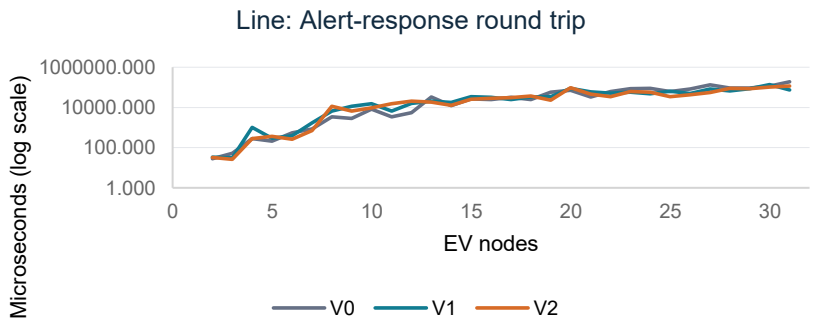
Broadcast/gather are collective elapsed times. ACK is sender completion. Every plotted category also has a detailed appendix graph.

Local synchronization and response times









Barrier: waiting for the round. Round trip: report send through advice receipt, including base processing.

CAAS measurements and implementation trade-offs

CAAS: 5 grid sizes + 5 lines x 3 variants x 3 launches = 90 scaling runs.

Grid query: V0/V1 = 1.64x; V0/V2 = 1.81x.

Line query: V0/V1 = 2.34x; V0/V2 = 2.49x.

Grid round trip: V0/V2 = 1.09x; line = 1.30x.

Highest observed round-trip score: V2 (1.197).

Ratios above 1 favour the refined version. Three launches; descriptive evidence.

Candidate compute at N=10,000: V0/V2 = 1709.9x (isolated).

Local and CAAS results are separate. No universal best variant is established.

Evidence expiry trades caution for recovery speed

CAAS expiry (s)	Alerts	Candidate recommendations
0	24	6
0.1	24	6
1	24	0
30	24	0

Six rounds; accelerated 0.05-second tick. Shorter expiry can forget congestion sooner. Missing evidence does not guarantee capacity.

Limitations: synthetic arrivals, global round barriers, inferred capacity and three repeats. Next: longer controlled runs, portable reports and candidate confirmation.

Questions

Thread safety and MPI matching

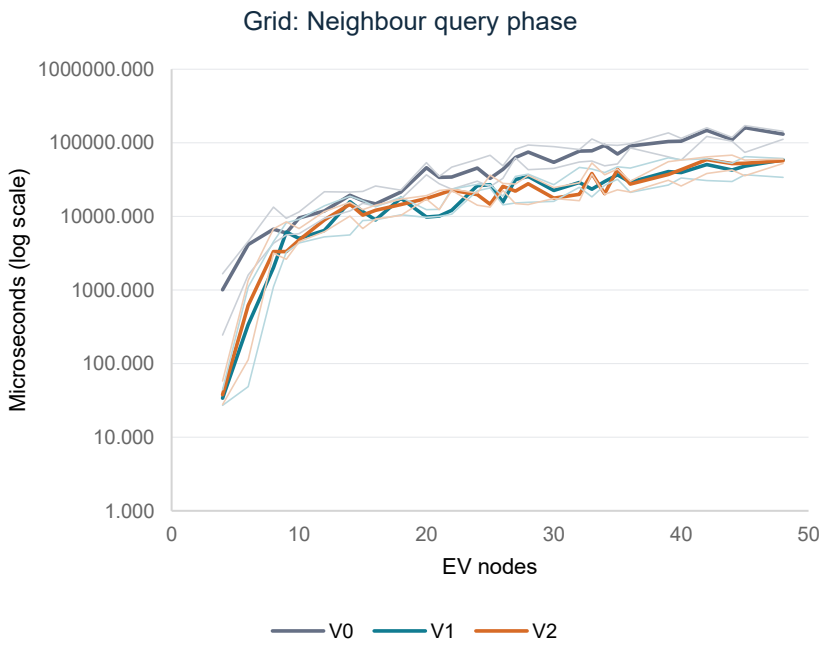
All ties, stale evidence and inferred availability

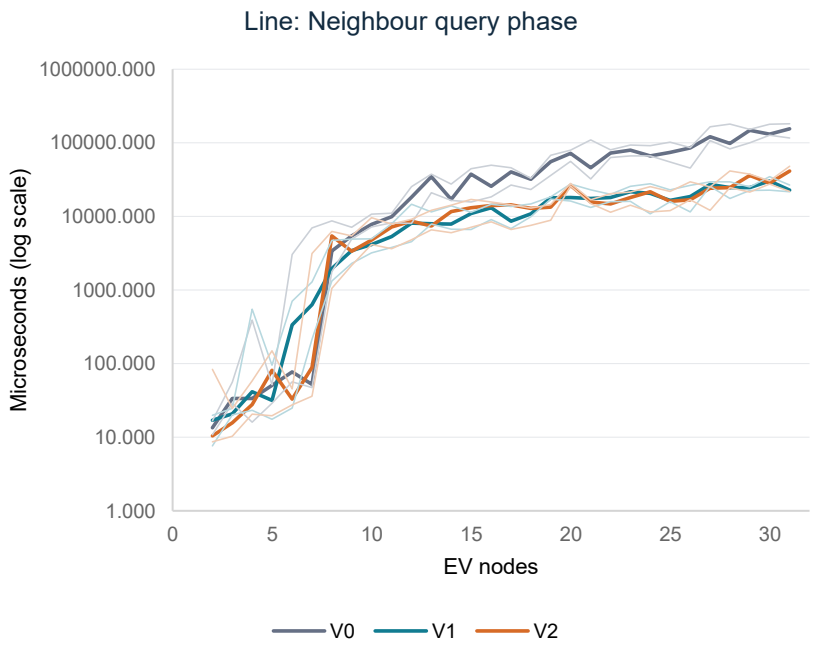
Communication costs and scaling limits

Shutdown: STOP broadcast, thread-team exit, completed requests, flushed logs, all acknowledgements, communicator cleanup and MPI_Finalize.

End of main presentation. The remaining pages are clearly labelled reference appendices.

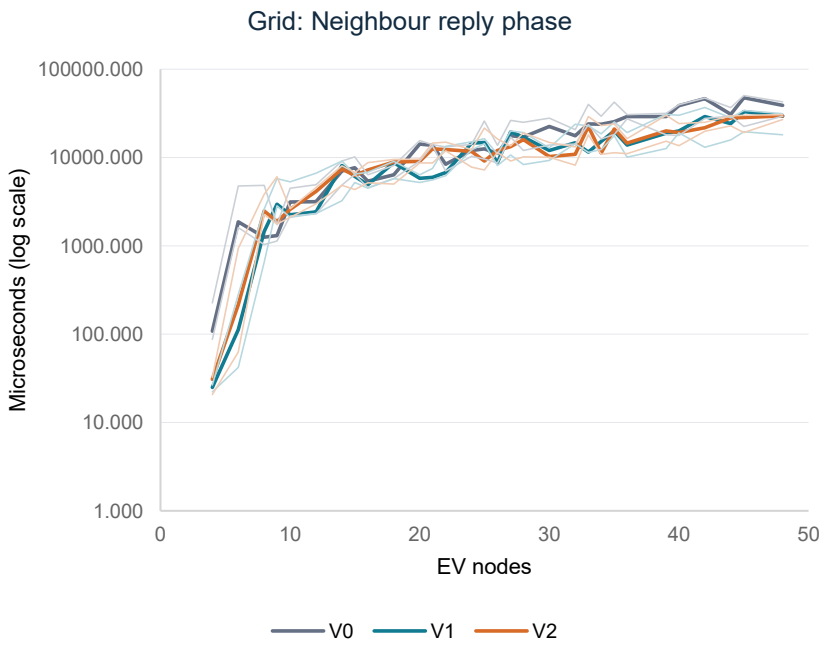
Neighbour query phase

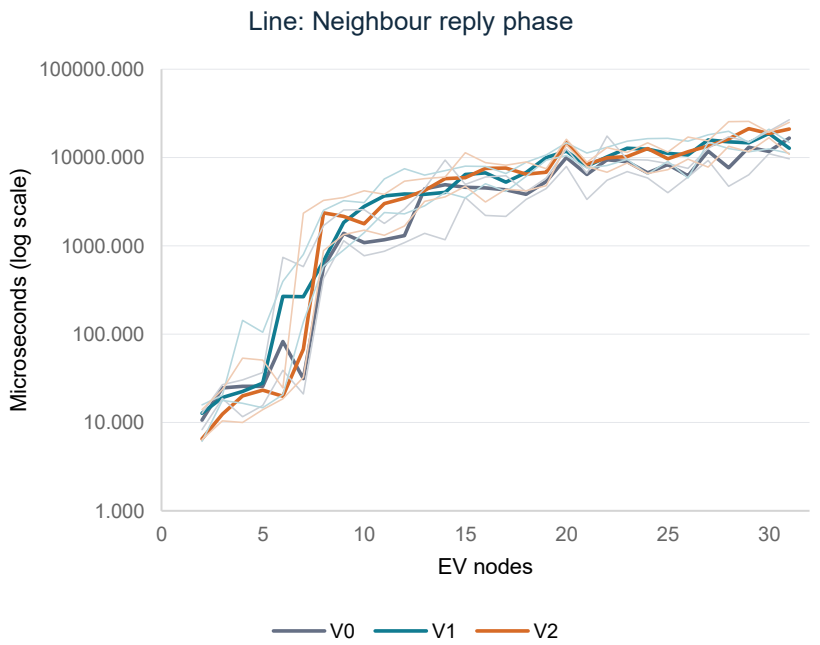




Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

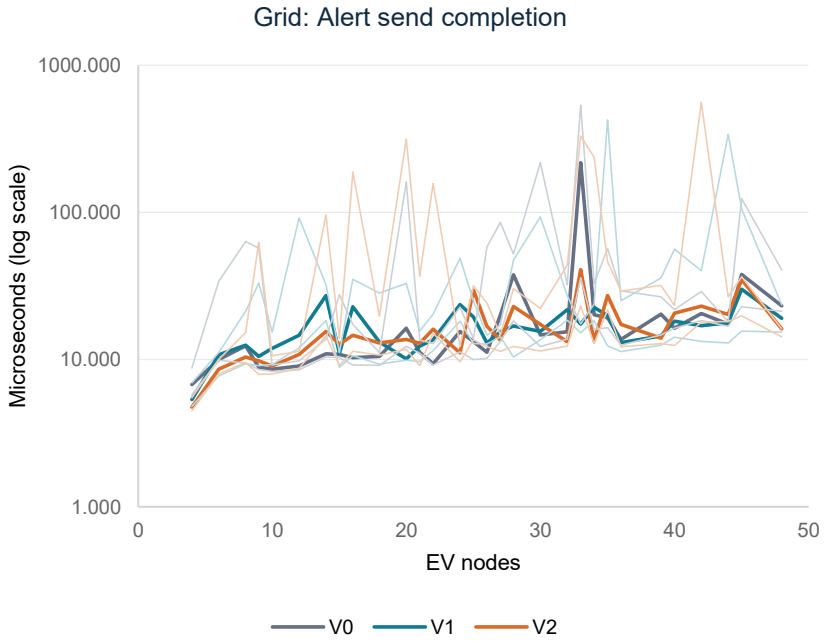
Neighbour reply phase

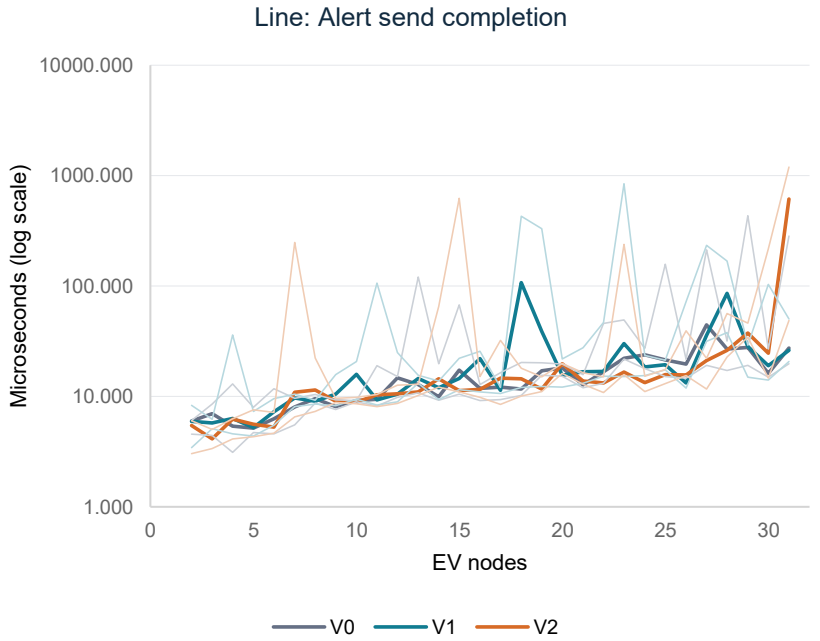




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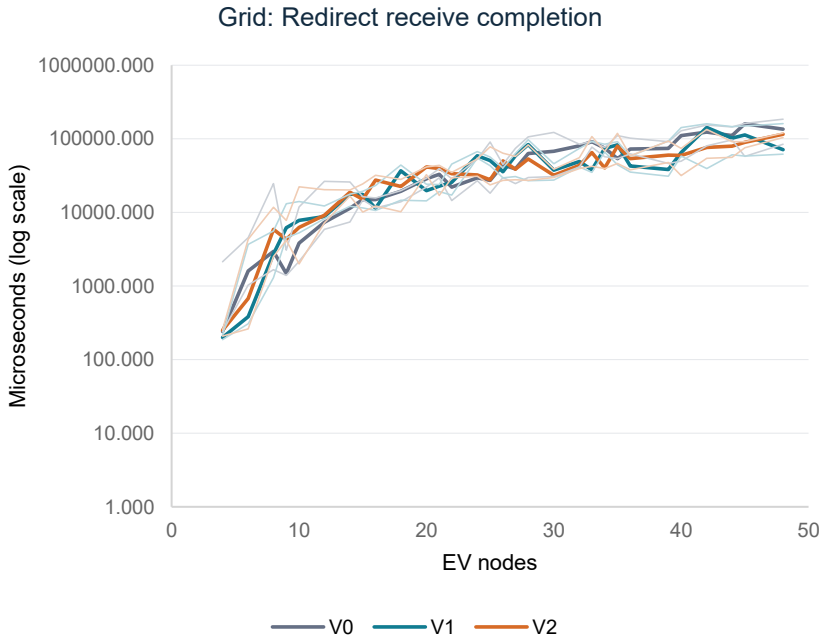
Alert send completion

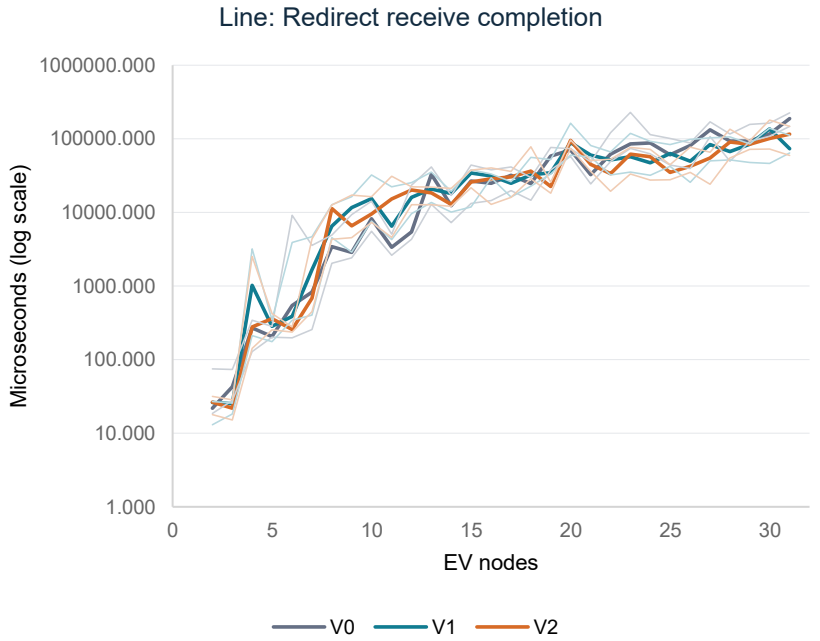




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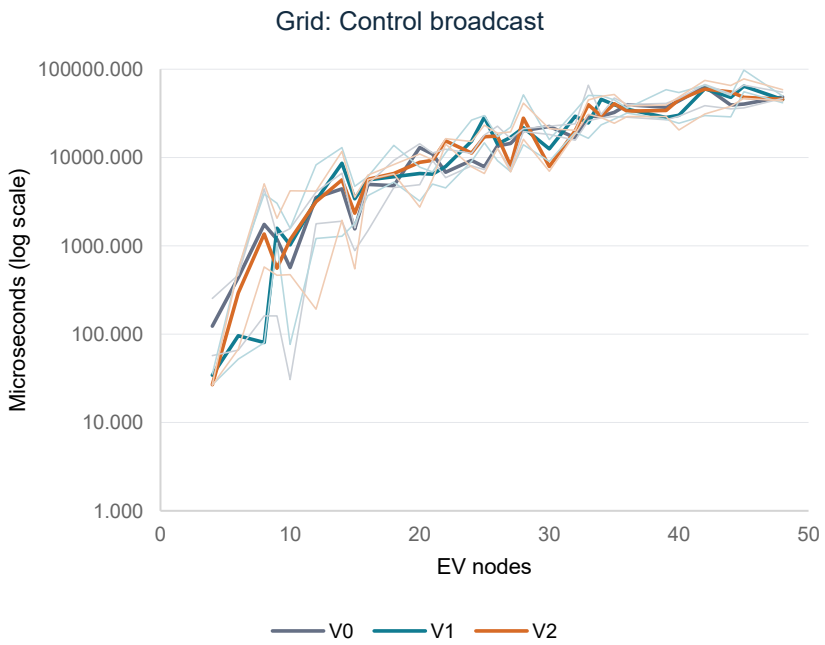
Redirect receive completion

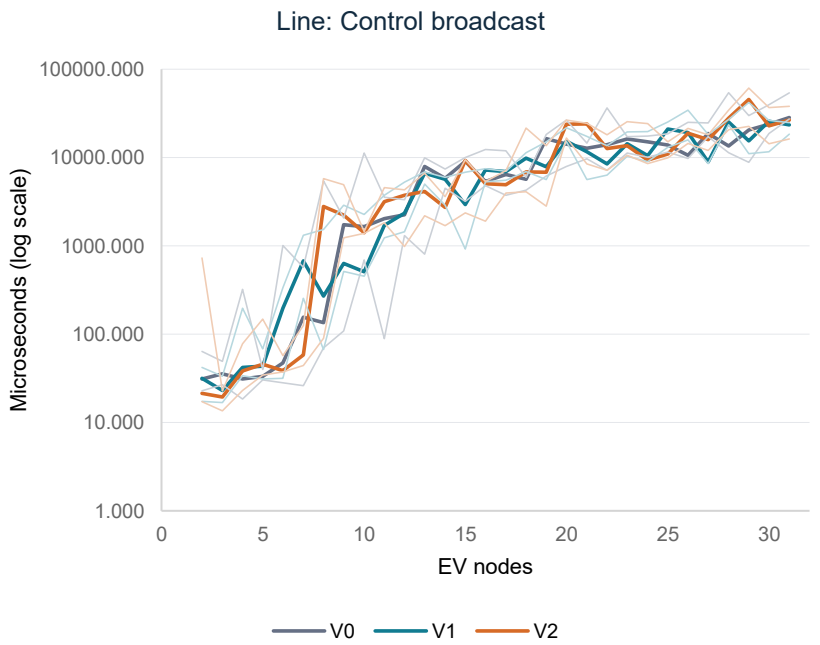




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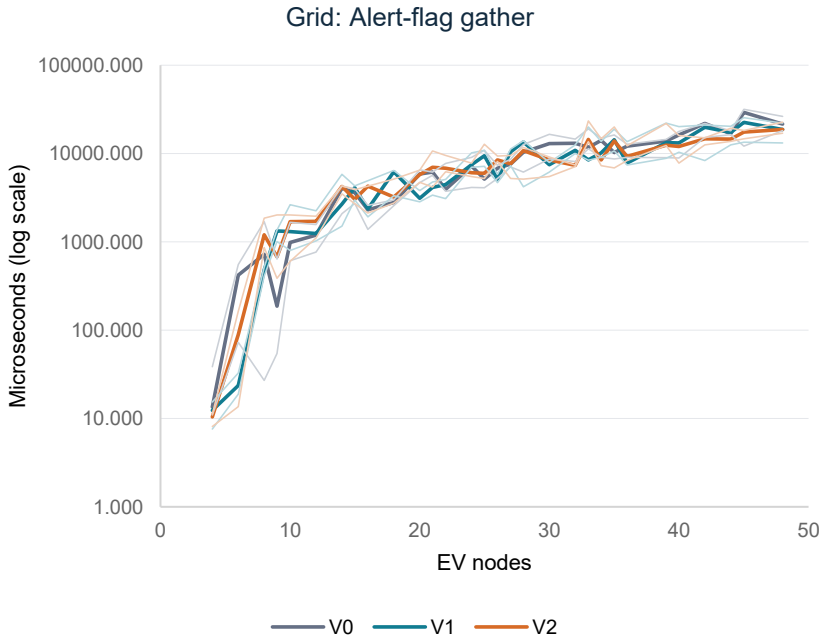
Control broadcast

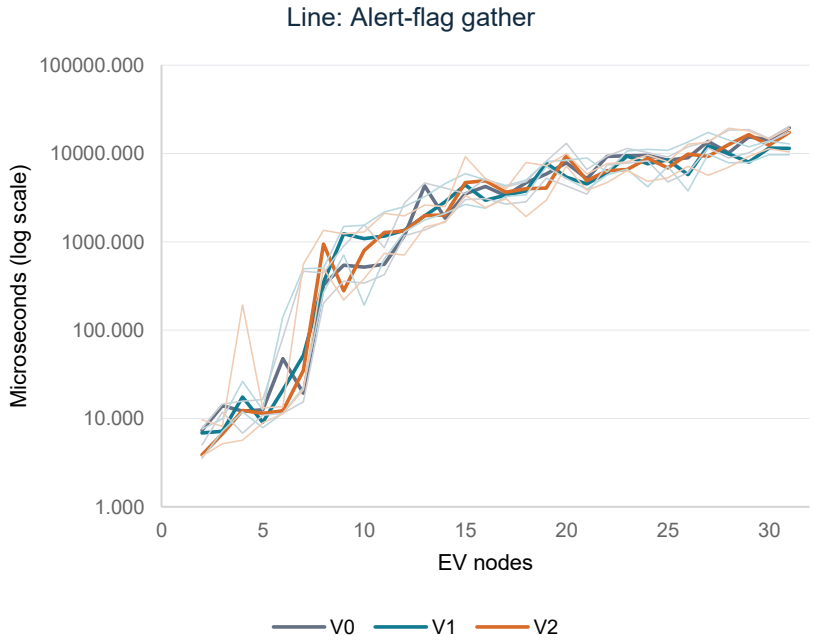




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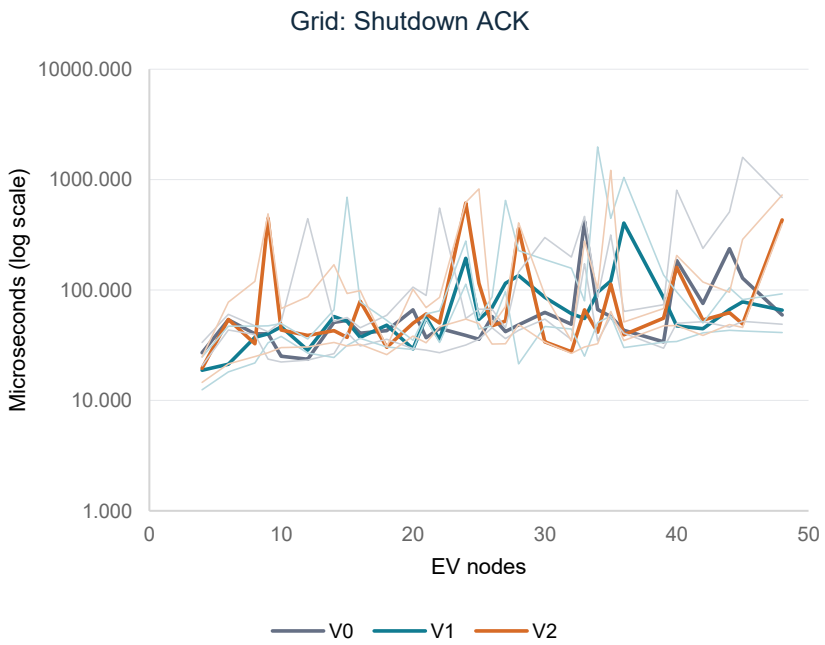
Alert-flag gather

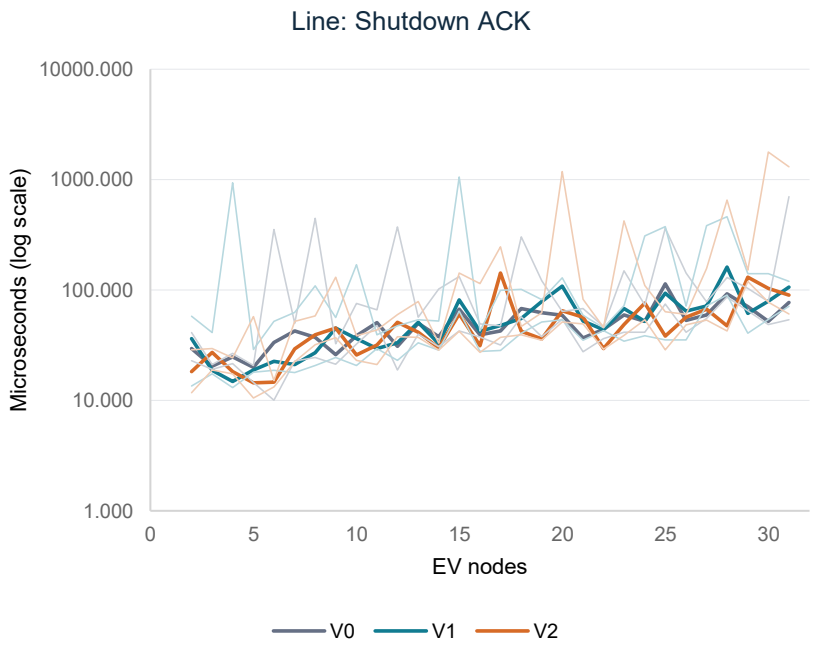




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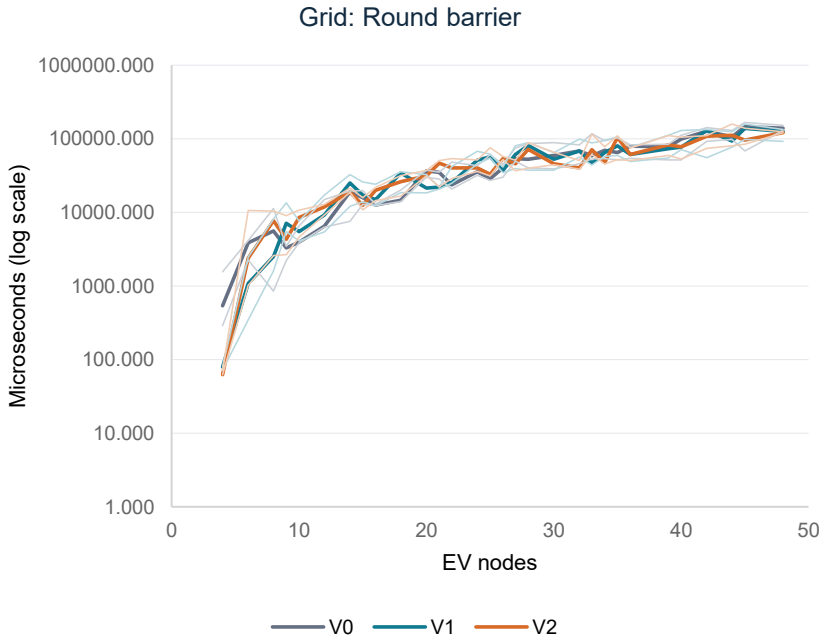
Shutdown ACK

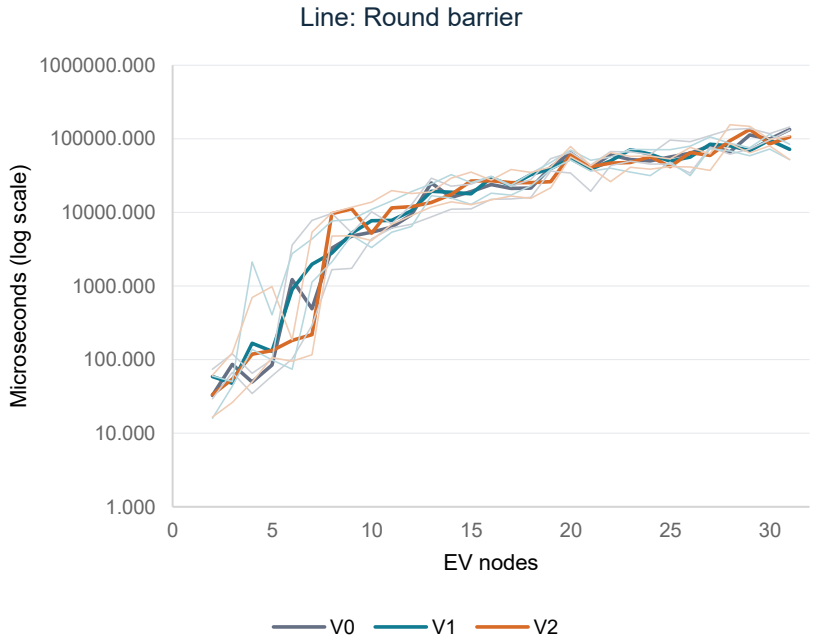




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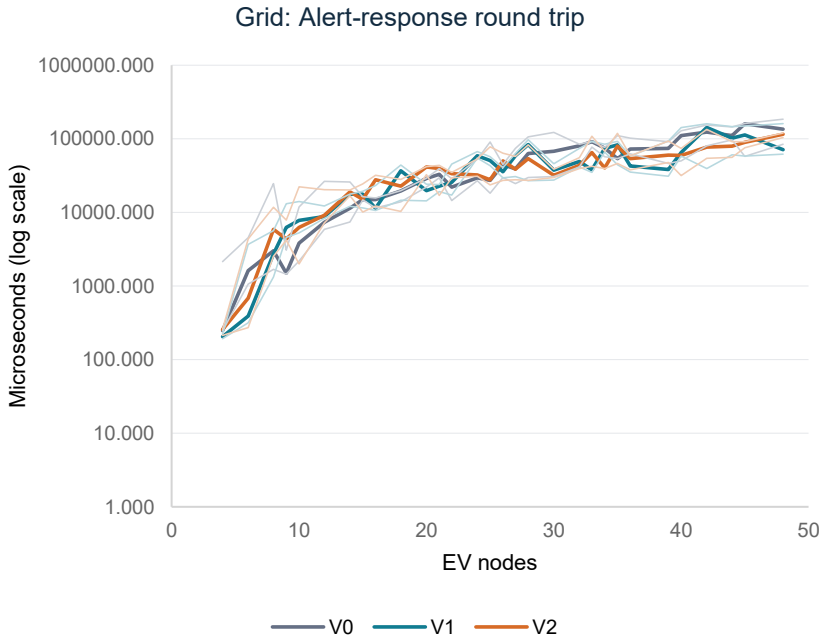
Round synchronization

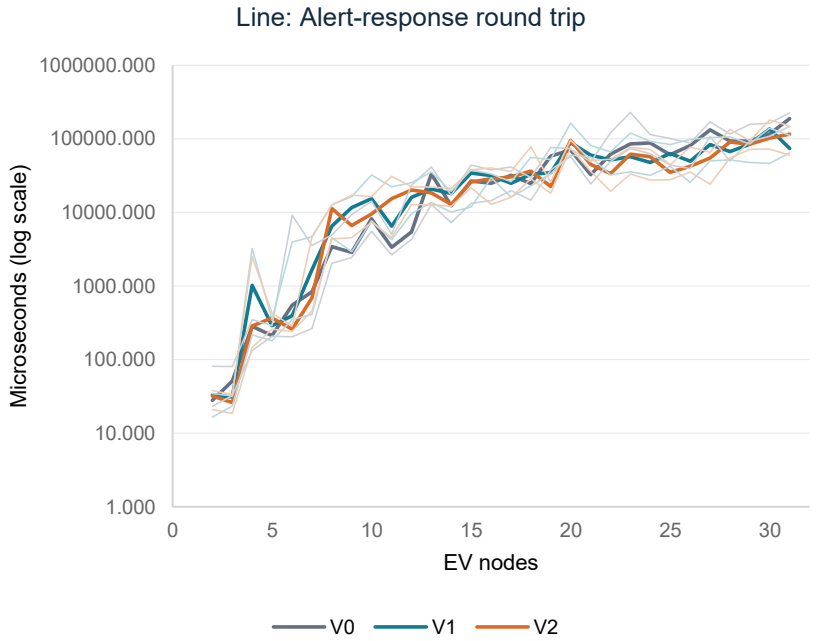




Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

Alert-response round trip





Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

Source and evidence index

Assessment specification, pages 5-8; rubric HD column, pages 1-5 (project docs/).

CAAS access guide, supplied demo scripts, caas_usage.txt and live defq limits.

Teaching context: WarmUpTask_1.c, WarmUpTask_2.c, MPI_VirtualTopologyExample.c.

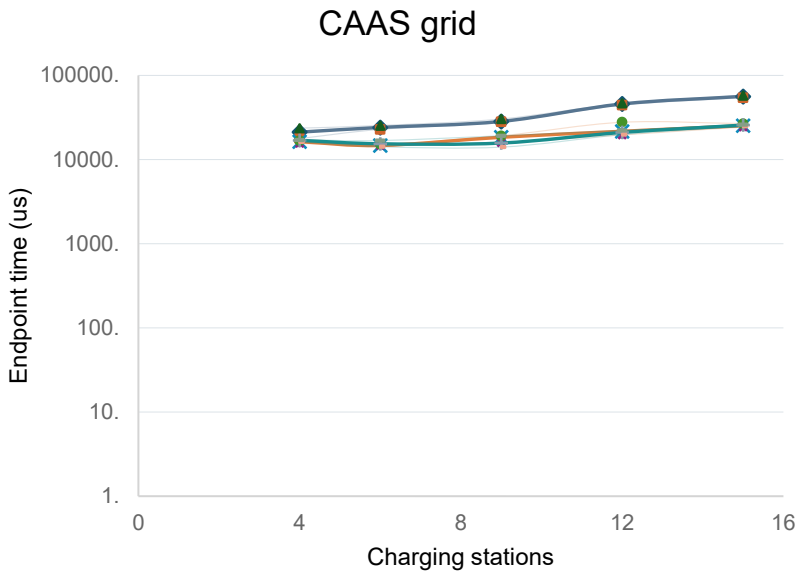
src/wsn.c; tests/; results/cluster/ and results/analysis/: verified local study.

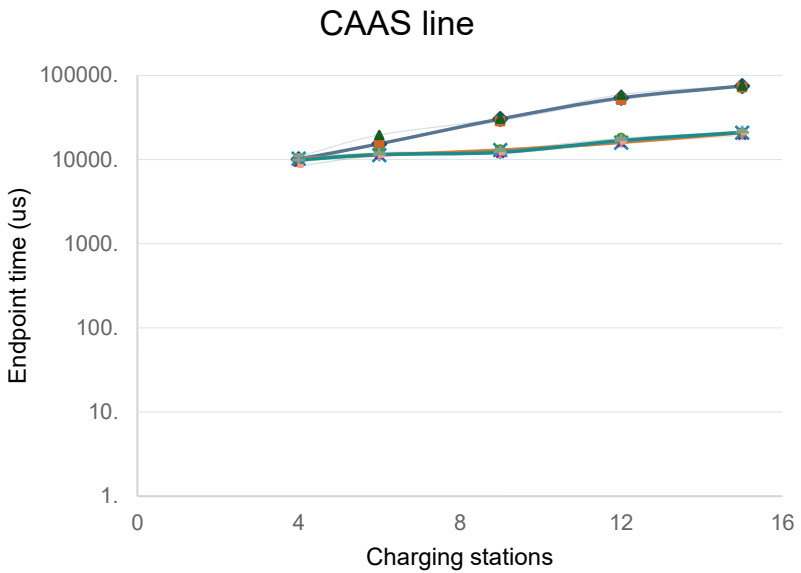
results/caas/: scheduler records, raw measurements, logs, placement and graphs.

Technical report: design, source hashes, limitations and AI declaration.

AI_Prompt_Records.pdf: supplied historical prompts and this audit request.

CAAS: Neighbour query phase

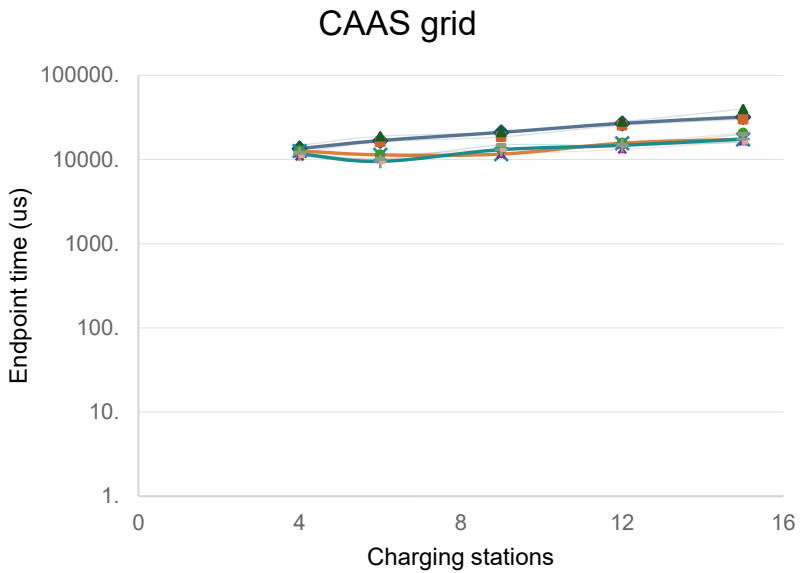


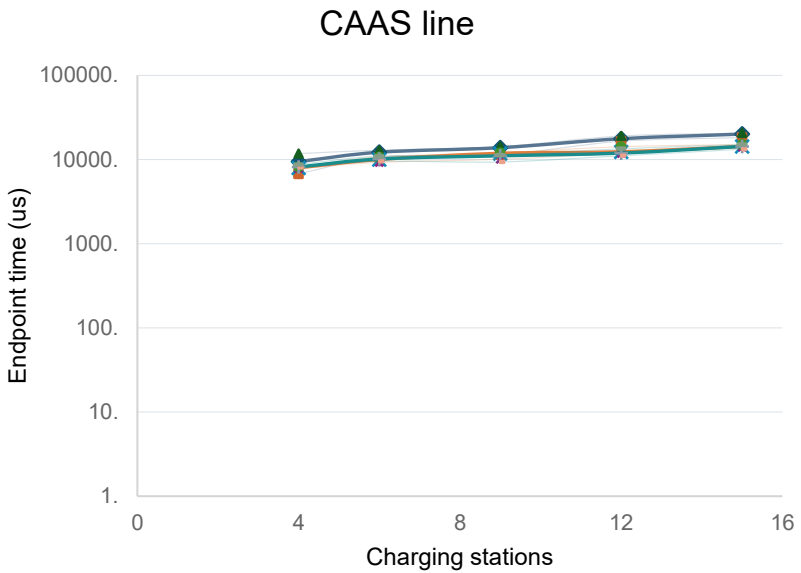


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per measured cycle. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Neighbour reply phase

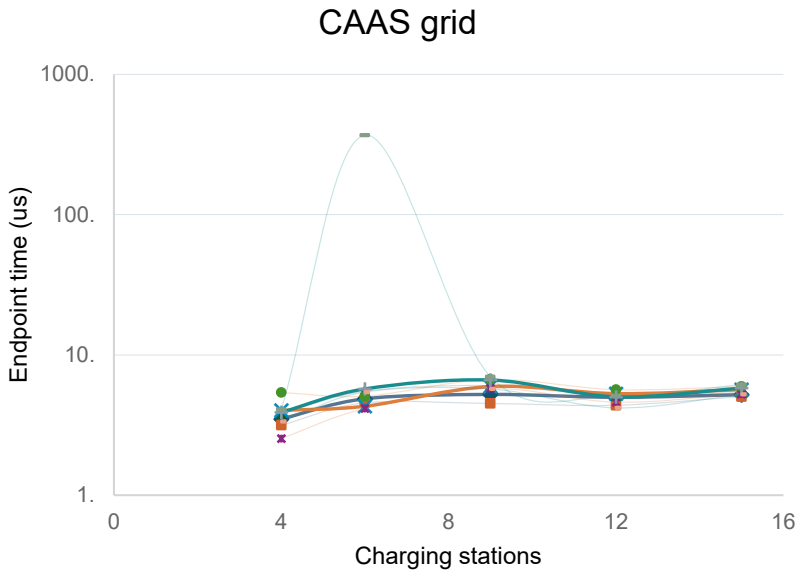


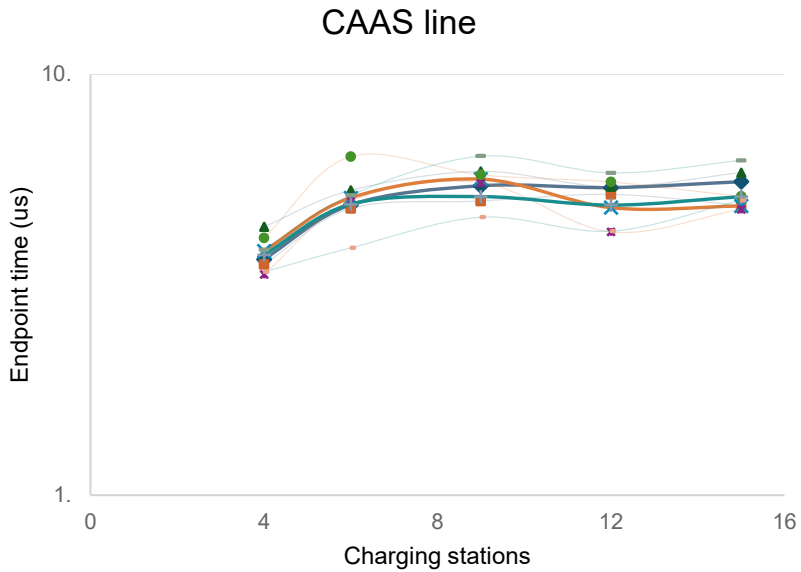


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per measured cycle. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert send completion

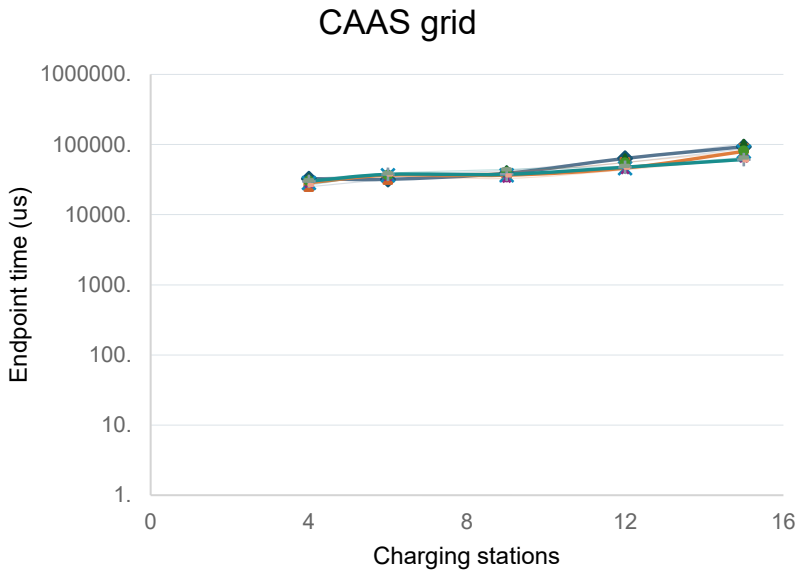


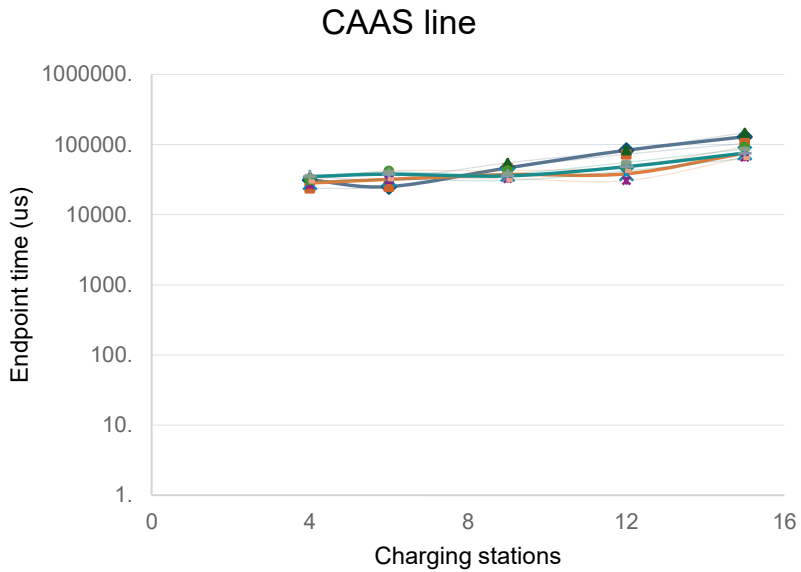


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Redirect receive completion

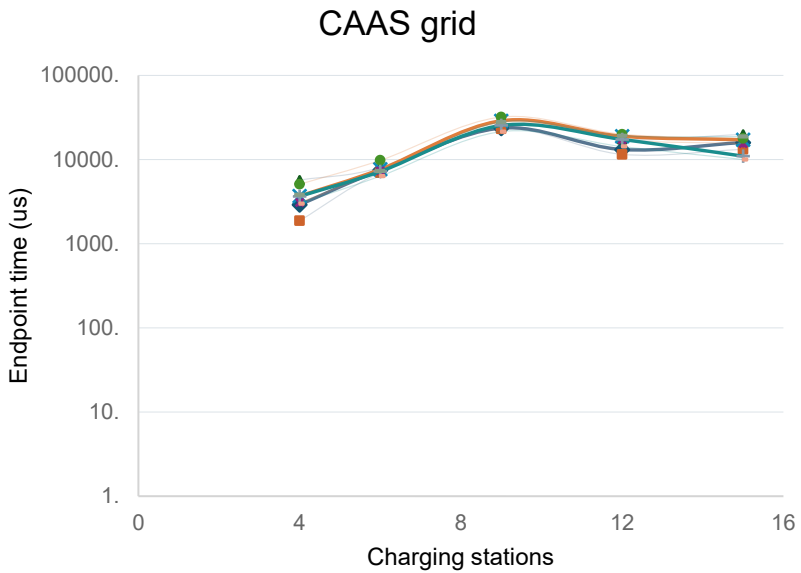


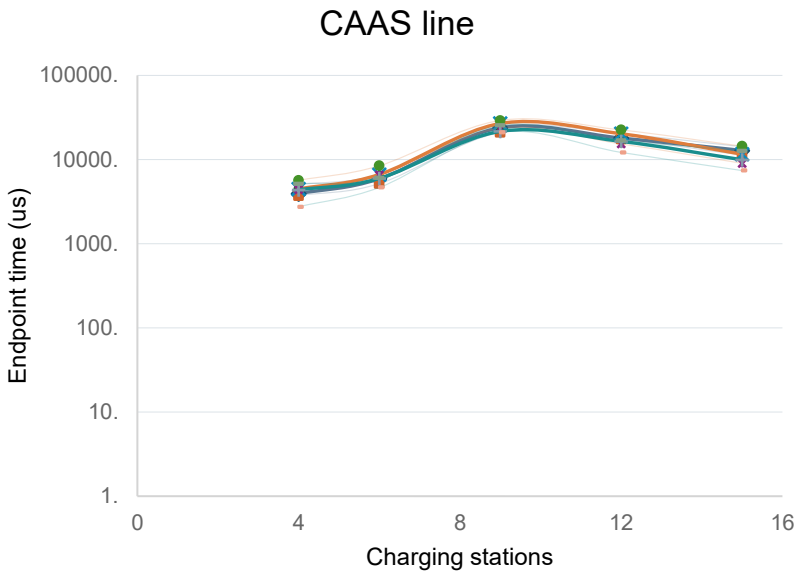


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Control broadcast

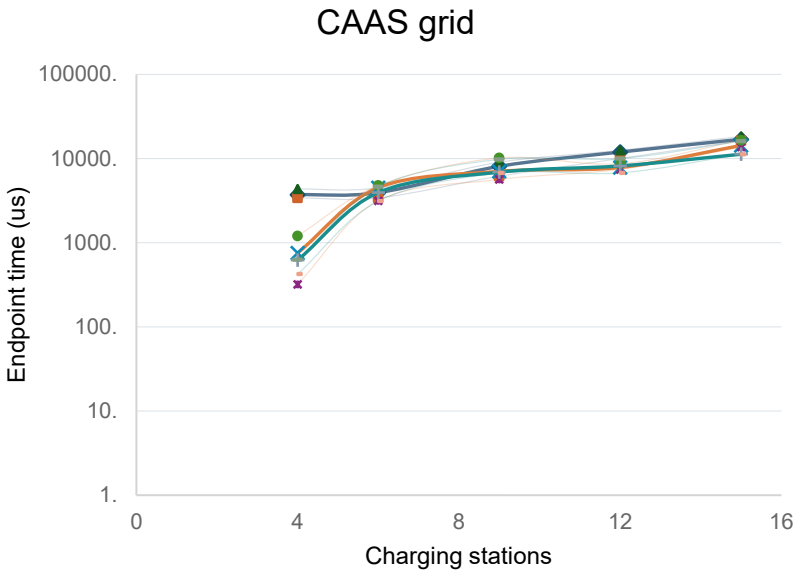


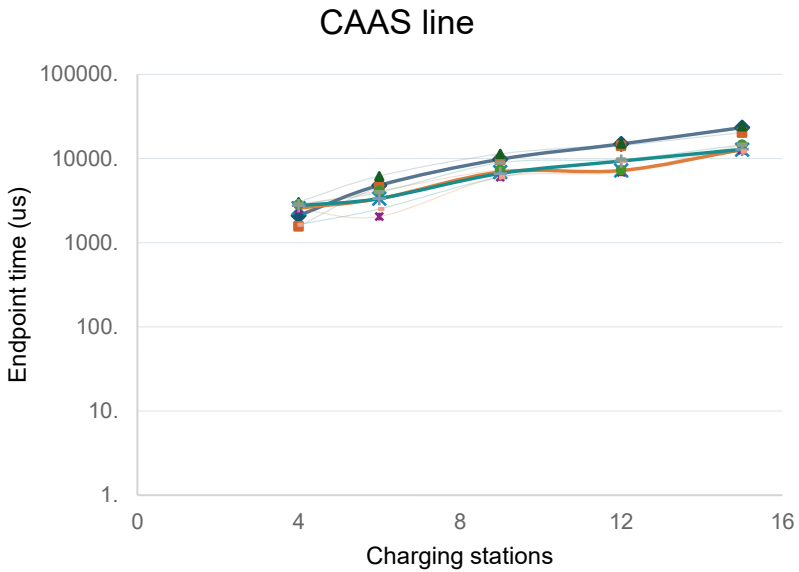


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert-flag gather

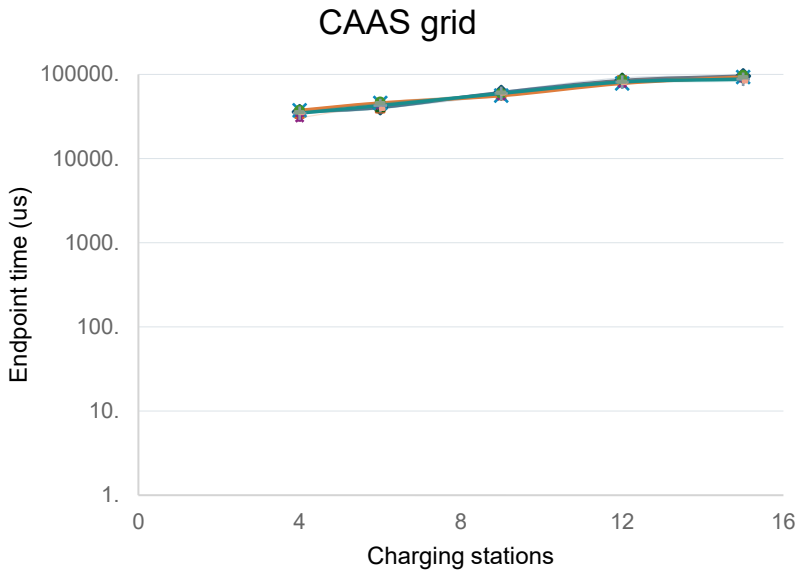


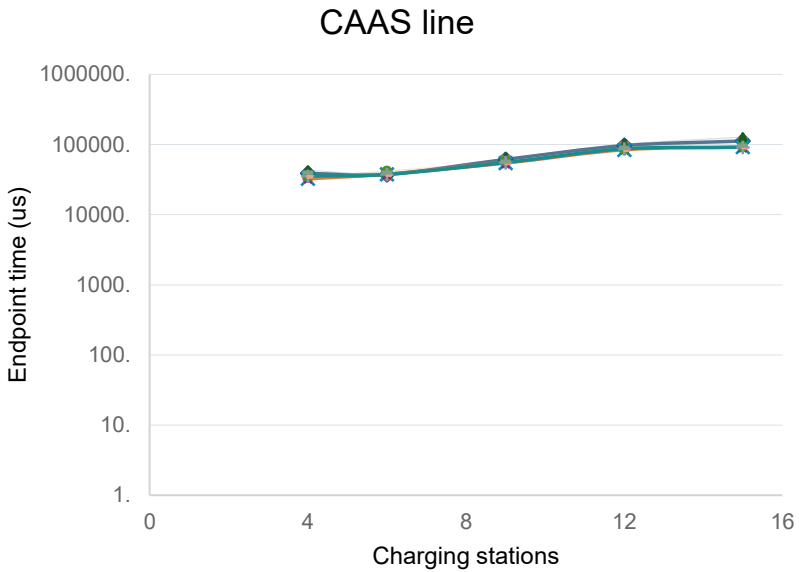


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Round synchronization

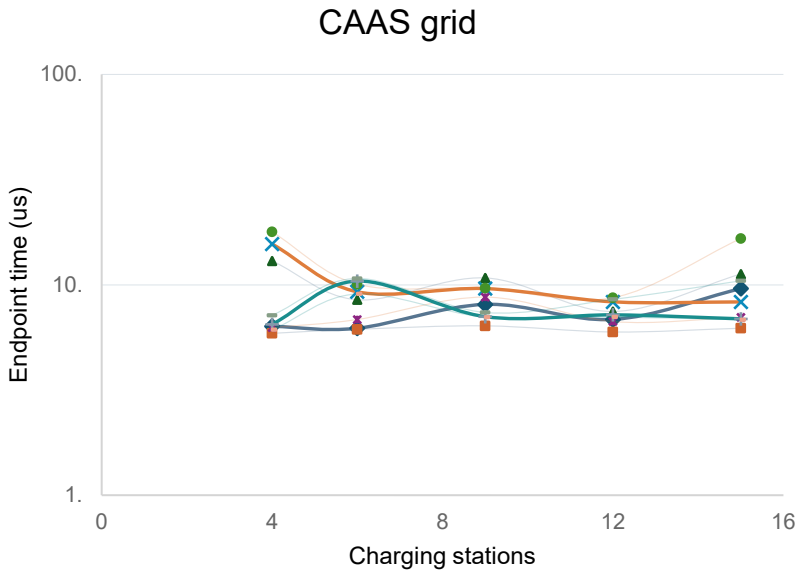


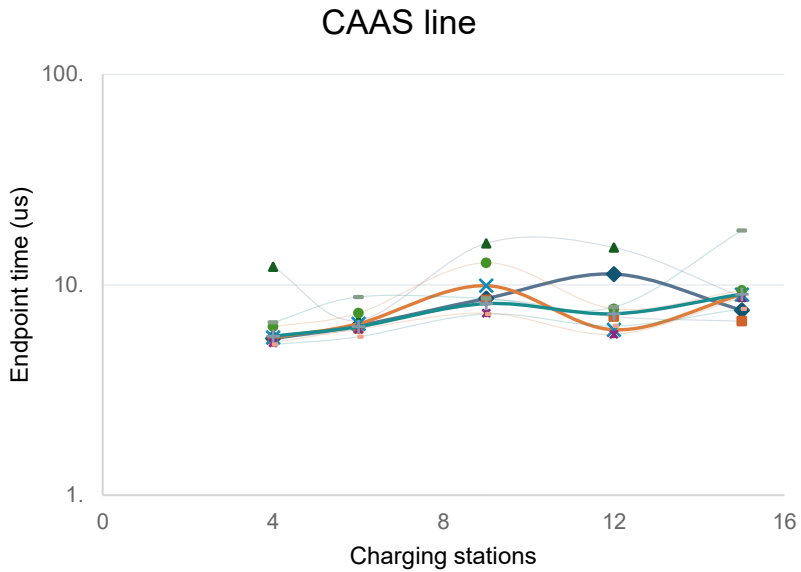


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Shutdown ACK

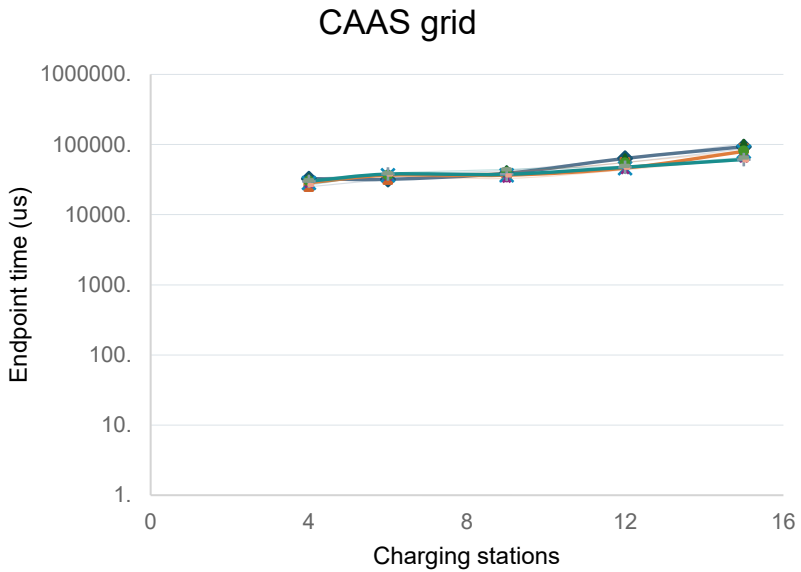


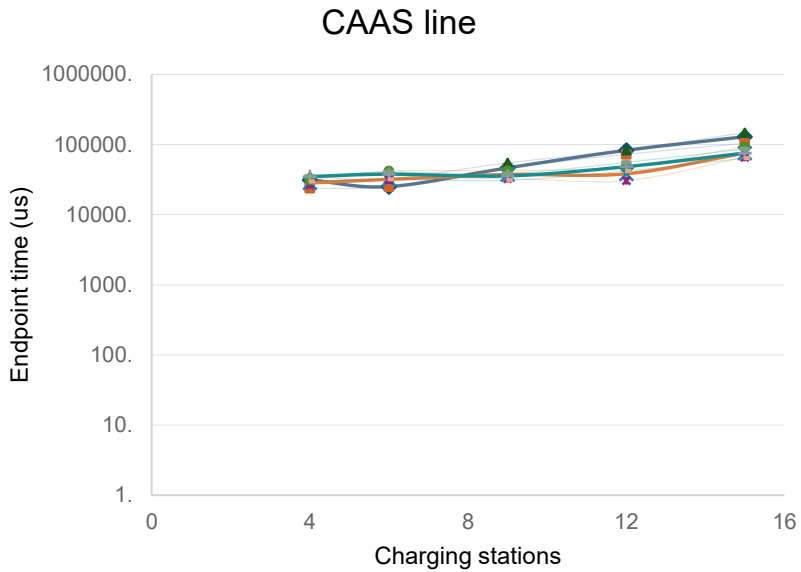


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert-response round trip





V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS completion and correctness evidence

CAAS job 199603: execution and completion

Local read-only evidence viewer. Content below comes from downloaded CAAS files; this is a browser view of those records.

SLURM accounting: selected original rows

```
JobID|JobName|State|ExitCode|Elapsed|NodeList|AllocCPUS
199603|fit3143-lab3-audit|COMPLETED|0:0|00:06:54|student-caas-n[10-11]|32
199603.batch|batch|COMPLETED|0:0|00:06:54|student-caas-n10|16
```

Correctness and completion: selected original stdout lines

```
PASS: ties, recent alert, recent low observation, expiry, isolated node
PASS: independent oracle for every station in all 1..7 by 1..7 shapes, four cache states, three variants
PASS 3x3 ties: all three versions; complete shutdown
PASS: 24 correctness runs, 90 scaling runs, 4 expiry runs, 1 realistic run, candidate tests and benchmark.
SCALING 90/90 line_1x15_v1_r2 4.949s
CAAS_EXPERIMENTS_COMPLETE
```

Source files

```
results/caas/job_accounting.txt
results/caas/slurm-199603.out
Full source logs and per-rank raw timings are retained in the submission package.
```

Authentic screenshot of the local evidence viewer displaying unmodified CAAS scheduler/test output.

CAAS all-ties and shutdown evidence

CAAS 3x3: all nearest ties and graceful shutdown

Local read-only evidence viewer. Content below comes from downloaded CAAS files; this is a browser view of those records.

Actual recorded launch argv

```
srun --mpi=none --exact --nodes=2 --ntasks=10 --ntasks-per-node=5 --cpus-per-task=2 --distribution=cyclic --cpu-bind=cores ./wsn_caas 3 3 2 --
variant 2 --scenario ties --rounds 4 --warmup 0 --tick 0 --expiry 30 --output results/caas/check_3x3_ties_v2
```

Original base log excerpt: header, first incident and shutdown summaries

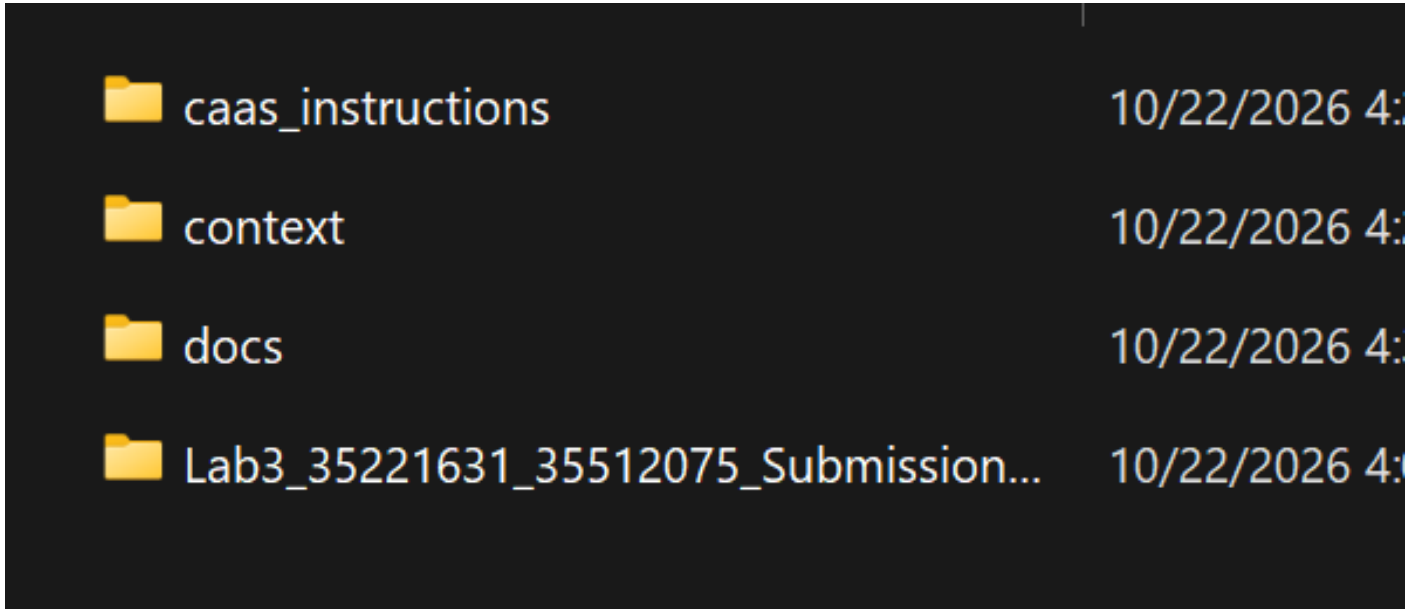
```
# host=student-caas-n10 world=0 station=-1 variant=2 ports=2 threshold=1 expiry=30 tick=0
{"round":0,"source":0,"coordinates":[0,0],"discovered_utc":"2026-10-
22T10:18:29Z","free":0,"neighbour_messages":8,"neighbour_bytes":32,"neighbours":[{"rank":-2,"free":-1},{"rank":3,"free":0},
{"rank":-2,"free":-1},{"rank":1,"free":0}],"inferred_candidates":[2,4,6]}
# alerts=4 elapsed_seconds=0.781598678 shutdown_acks=9
# measured_point_to_point_messages=145 measured_payload_bytes=900 collective_operations=13 (warmup excluded; MPI protocol overhead excluded)
```

Source record

```
CAAS job 199603
Raw file: check_3x3_ties_v2.base0.log
The complete incident sequence is retained in all_base_station_logs.log.
```

Authentic screenshot of the local evidence viewer displaying a CAAS 3x3 all-ties base log.

Supplied project folder structure



User-supplied local folder structure: the assessment inputs and original implementation package.

This image records file organisation only. CAAS execution is evidenced by the separate verified logs and captures.